SpecTrack: Spectral-Heterogeneous MoE for Multispectral Object Tracking

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Abstract

Multispectral object tracking leverages spectral signatures to robustly handle camouflage and background clutter. However, current methods rely on an inefficient uniform processing paradigm that treats all regions equally. This approach squanders resources on redundant backgrounds and lacks the adaptive capacity needed for discriminatively critical regions, such as object boundaries and spectrally similar distractors. To address these limitations, we propose SpecTrack, a novel framework that dynamically aligns model capacity with the local spectral-spatial complexity of the scene. The core is the Spectral-Heterogeneous Mixture of Experts (SHMoE), which adaptively reallocates computational budget through a set of specialized experts. Also, We introduce a Frequency Spectral Prompt Router that integrates spatial semantics with frequency aware spectral cues to identify hard regions, directing them to capacity heavy experts for intensive modeling while assigning lightweight experts to easy backgrounds. Experimental results show that SpecTrack outperforms state-of-the-art trackers by 5.5% and 4.1% AUC on MUST and MSITrack datasets, respectively while also demonstrating robust generalization ability on the large-scale RGB benchmark GOT-10k. For flexible deployment, we introduce five variants catering to devices from edge to GPUs, striking a good trade-off between speed and accuracy.

1 Introduction

Visual **S**ingle **O**bject **T**racking (SOT) serves as a cornerstone in computer vision systems [Li *et al.*, 2025; Chen *et al.*, 2022], underpinning critical applications ranging from UAV surveillance [Kou *et al.*, 2025] to autonomous driving [Yang *et al.*, 2023]. While RGB-based trackers have witnessed remarkable progress [Ye *et al.*, 2022; Li *et al.*, 2025], their reliance on visible bands often renders them fragile in scenarios involving background clutter or camouflage [Wang *et al.*, 2024]. In such cases, visual appearance alone is insufficient for robust discrimination. Multispectral Imaging (MSI) emerges as a compelling alternative by capturing material-dependent reflectance across visible and near-infrared bands [Xiong *et al.*, 2020a]. As illustrated in Fig. 1, distinct spectral signatures reveal the intrinsic physical properties of objects, enabling reliable tracking even when the target and background exhibit indistinguishable colors or textures [Qin *et al.*, 2025; Feng *et al.*, 2025].

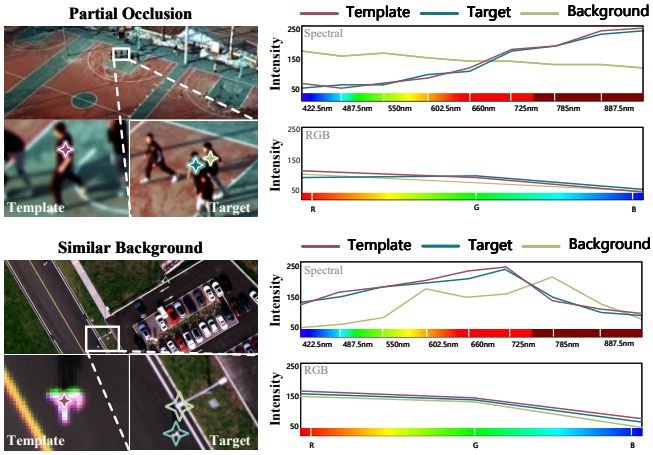

Figure 1: In challenging scenarios, the spatial features (e.g., color and texture) of the tracked target closely resemble those of the background, making differentiation and localization difficult. However, the target's spectral information differs significantly from the background and aligns with the template's spectral data, providing robust features for reliable tracking.

Despite the promise of MSI, existing multispectral object tracking methods largely follow a uniform processing paradigm, employing identical feature extraction and relation modeling strategies across all image regions. The paradigm persists across various architectures, including ensemble frameworks [Li *et al.*, 2023; Li *et al.*, 2020; Li *et al.*, 2021], Siamese architectures [Liu *et al.*, 2022; Chen *et al.*, 2023c] and unified Transformers networks [Qin *et al.*, 2025]. Existing methods fail to account for the inherent imbalance in discriminative difficulty across the scene.

In tracking scenarios, vast background regions often exhibit semantic redundancy. Conversely, a small subset of discriminatively critical regions, including object boundaries and spectrally similar distractors, exhibits severe ambiguity.

Unlike the redundant background, these regions demand deep reasoning. Applying a uniform processing strategy to such diverse regions is inefficient. On the one hand, allocating heavy computational resources to **easy** background tokens incurs unnecessary redundancy, limiting the model's efficiency. On the other hand, simple interaction mechanisms shared globally may fail to disentangle the complex mixed pixel signatures in hard regions, leading to diluted discriminative features. These limitations motivate our question: **How to design a tracker that adaptively allocates representational capacity based on the local spectral-spatial complexity of the scene?**

To address this question, we propose SpecTrack, a novel multispectral tracking framework that explicitly accounts for spectral heterogeneity. Instead of previous methods that rely on static architectures, SpecTrack dynamically adjusts its processing strategy through a Spectral Heterogeneous Mixture of Experts (SHMoE). Inspired by the success of MoE [Aljundi *et al.*, 2017; Tang *et al.*, 2024] in handling diverse data patterns, we employs a set of specialized experts with varying receptive fields and interaction granularities. Specifically, lightweight experts efficiently process confident background regions to conserve resources, while capacity-heavy experts concentrate computation on resolving ambiguity in hard foreground tokens and distractors.Furthermore, to prevent the fragmentation of global context caused by sparse expert activation, we incorporate a spectral-domain global interaction module to maintain long-range information propagation. To ensure precise expert assignment, we design a Frequency Spectral Prompt Router. Instead of merely detecting high-frequency edges, this router integrates spatial semantics with frequency-aware spectral cues to identify discriminatively hard regions that require intensive modeling.

Benefiting from the adaptive resource allocation of SHMoE, SpecTrack achieves a superior trade-off between tracking robustness and computational efficiency. Contrary to dense models that process all tokens with equal complexity, our method focuses representational capacity on high value regions, effectively suppressing background noise while sharpening target features. We validate the effectiveness of SpecTrack through extensive experiments on four diverse benchmarks. Empirical results demonstrate that our framework sets new state-of-the-art standards across multispectral datasets (MUST [Qin *et al.*, 2025], MSITrack [Feng *et al.*, 2025]) and the hyperspectral benchmark (HOT2022 [Xiong *et al.*, 2020a]). Furthermore, we extend our evaluation to the large-scale RGB dataset GOT-10k [Huang *et al.*, 2021], verifying the scalability and generalization capability of our tracker in generic scenarios. As shown in Tab. 2, Tab. 3, Tab. 4 and Tab. 5, SpecTrack-B224 achieves or surpasses the performance of previous state-of-the-art trackers while maintaining comparable inference speeds.

Our contributions are summarized as follows: (1) We identify the inefficiency of uniform processing in MSI tracking and propose SpecTrack, the first framework to introduce MoE for adaptive spectral-spatial modeling. (2) We design the SHMoE module equipped with a Frequency Spectral Prompt Router, which effectively decouples simple backgrounds from complex foregrounds, aligning model capacity with region-specific demands. (3) Extensive experiments on three multispectral benchmarks demonstrate that SpecTrack consistently outperforms state-of-the-art trackers, validating the effectiveness of our adaptive modeling strategy.

2 Related Work

2.1 Multispectral Object Tracking

Unlike standard RGB tracking, Multispectral Object Tracking leverages multiple spectral bands to improve target discrimination under challenging conditions such as low illumination and background camouflage. Hyperspectral imagery provides rich material-specific cues through tens to hundreds of narrow-band channels, enabling the detection of camouflaged or low-contrast targets. This has driven the development of dedicated multispectral tracking benchmarks, including HOT2022 [Xiong *et al.*, 2020b], MUST and MSITrack.

Early multispectral trackers relied on hand-crafted spectral features or correlation filter-based frameworks [Xiong *et al.*, 2020a]. With the advent of deep learning, ensemble-based methods such as BAE-Netand SEE-Net [Li *et al.*, 2023] improved performance by processing grouped pseudo-color representations with attention or learned band importance. More recently, SiamHYPER [Liu *et al.*, 2022] adapts Siamese architectures to multispectral data via knowledge transfer, while UNTrack [Qin *et al.*, 2025] employs a unified Transformer backbone to integrate spectral prompt with spatial features. Despite these advances, existing multispectral trackers generally adopt static computational paradigms, uniformly processing all regions of the hyperspectral volume. This leads to substantial redundancy in low-complexity regions and limits the prioritization of spectrally challenging targets. In contrast, our SpecTrack dynamically allocates computation according to spectral–spatial complexity.

2.2 Mixture-of-Experts in Vision and Tracking

MoE[Jacobs *et al.*, 1991; Aljundi *et al.*, 2017] are widely applied in large language models[Puigcerver *et al.*, 2023; Wu *et al.*, 2024] , where each expert focuses on specific aspects of the data or particular tasks. By performing an adaptive weighted sum over multiple experts, MoE can better capture complex patterns and relationships within the input data. Inspired by these advances, recent studies have begun to explore MoE architectures in computer vision, with a growing focus on object tracking. MoETrack [Tang *et al.*, 2024] employs multiple prediction heads as experts in the field of RGB-T tracking, which is rather far-fetched and deviates from the general practice of embedding MoE within Transformer layers.Cai et al. introduce SPMTrack[Cai *et al.*, 2025], a transformer tracker with a TMoE module that routes spatial tokens to multiple experts for diverse relation modeling. Similarly, Sun et al. propose HotMoE[Sun *et al.*, 2025] for multispectral object tracking, which activates the top-k experts for each token in a sparse gating scheme. These works show that MoE can increase representational capacity with little added cost, but they still use homogeneous experts and rely only on spatial cues for routing. In contrast, our method explores a spectral-aware MoE design to better adapt computation to the characteristics of multispectral tracking.

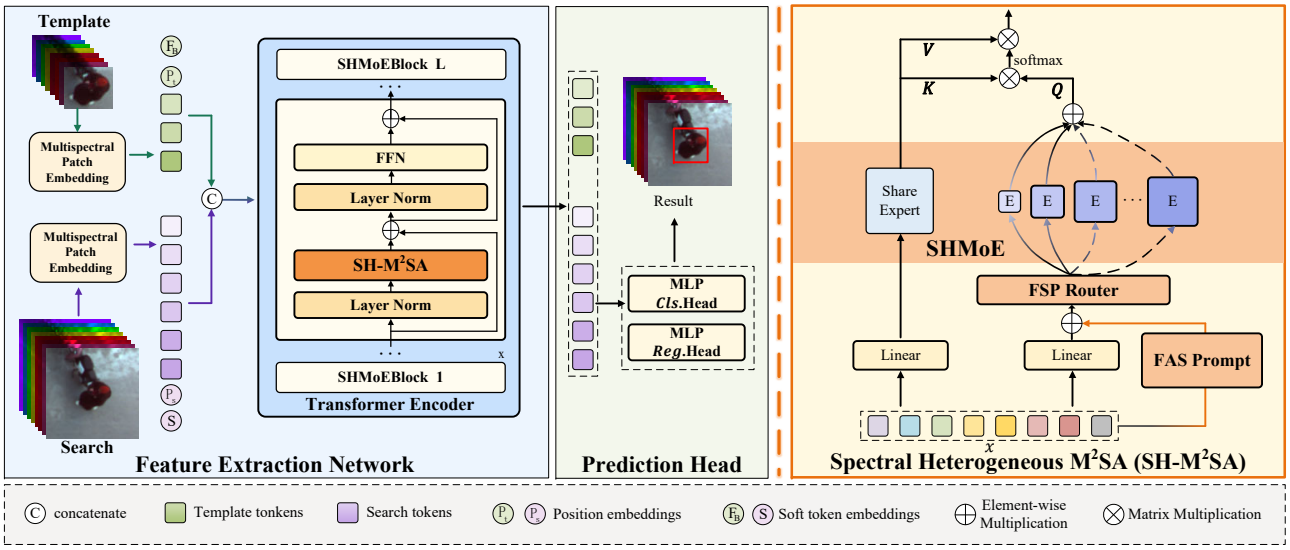

Figure 2: The Overview of the proposed SpecTrack framework that consists of a feature extraction network and a prediction head. The Transformer encoder is composed of stacked SHMoEBlocks, featuring SHMoE-based Multi-Head Self-Attention (SH-M^{2}SA). The structure of SH-M^{2}SA module is shown at the right side.

3 Method

3.1 Framework Overview

As illustrated in Fig. 2, we propose SpecTrack, a multi-spectral object tracker that explicitly exploits spectral heterogeneity through a Spectral Heterogeneous Mixture of Experts (SHMoE) design. The overall architecture follows the one-stream tracking paradigm and consists of a feature extraction network built upon OSTrack [Ye *et al.*, 2022] and a lightweight prediction head.

Feature Extraction and Embedding. The network accepts a single template image $Z \in \mathbb{R}^{H_z \times W_z \times C}$ from the first frame and a search region $S \in \mathbb{R}^{H_s \times W_s \times C}$ from the current frame, where C denotes the number of spectral bands. Unlike RGB trackers, the inputs contain rich multispectral information. To process this, the multispectral input is partitioned into patches and projected into a shared embedding space:

$$E_i = \mathrm{PE}(P_i), \quad P_i \in \mathbb{R}^{P \times P \times C}, \tag{1}$$

where $\mathrm{PE}(\cdot)$ represents a convolutional projection layer that maps raw multispectral patches into a unified latent embedding space. Subsequently, the template and search embeddings are concatenated, supplemented with learnable positional encodings, and input to the feature extraction network Φ.

SHMoEBlock Architecture. The core of the feature extraction network is a Transformer encoder composed of stacked SHMoEBlocks. As illustrated in the right panel of Fig. 2, the SHMoEBlock improves upon the standard vision Transformer block by replacing the standard multi-head self-attention (MSA) with the SHMoE-based MSA (SH-M^{2}SA) layer. To orchestrate feature processing, we employ the Frequency Spectral Prompt Router (FSP-Router). Instead of relying solely on spatial semantics which may be ambiguous in camouflaged scenarios, the FSP-Router incorporates high-frequency spectral cues extracted via fixed high-pass filters. These cues guide the model to accurately identify discriminatively hard regions (e.g., targets with similar textures to the background) and route them to experts with appropriate receptive fields. Formally, in the l-th SHMoEBlock, the input feature $\mathbf{Z}^{l-1}$ is processed as follows:

$$\hat{\mathbf{Z}}^l = \mathbf{Z}^{l-1} + \text{SH-M}^2\text{SA}(\text{LN}(\mathbf{Z}^{l-1})), \tag{2}$$

$$\mathbf{Z}^l = \hat{\mathbf{Z}}^l + \text{FFN}(\text{LN}(\hat{\mathbf{Z}}^l)). \tag{3}$$

By stacking L such blocks, the model progressively extracts discriminative multispectral features.

Prediction Head. The prediction head receives the search region tokens from the final SHMoEBlock and reshapes them to restore the 2D spatial structure. We adopt the traditional center point prediction method [Ye *et al.*, 2022], where the regression and classification branches predict the target bounding box based on the refined features.

3.2 Spectral Heterogeneous Mixture of Experts

Standard MoE architectures typically employ homogeneous experts with identical architectures. While this simplifies implementation, it lacks adaptability to the heterogeneity of complexity inherent in multispectral data. We address this by designing a set of experts that share a unified FFT-based architecture but differ in their receptive field configurations to match varying processing needs.

Nested Expert Design. We construct a set of experts $\mathcal{E}$ that are differentiated by window size w and network depth D. While all experts utilize the FFT-Attention mechanism for efficient mixing, they are parameterized differently:

- **Lightweight Experts (E_{light}):** These experts utilize shallow networks ($D = 1$) with small receptive fields.

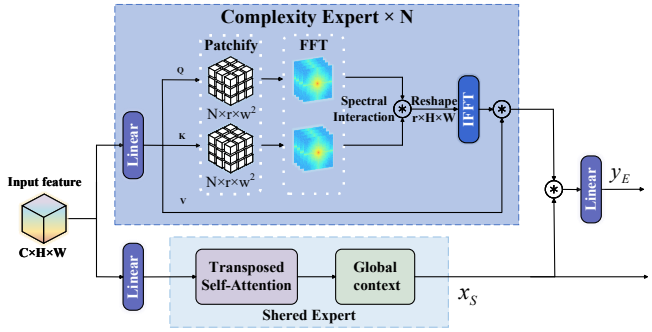

Figure 3: Illustration of the Spectral-Domain Global Interaction mechanism. It synergizes local frequency-domain features from the Complexity Expert with global context from the Shared Expert. This interaction prevents feature fragmentation by ensuring that local spectral processing is enriched with holistic information.

They are designed to efficiently process simple background regions (e.g., sky, road) where local spatial aggregation is sufficient, thereby conserving computational resources.

- **Heavyweight Experts (E_{heavy}):** These experts utilize deeper layers ($D \geq 2$) and are equipped with larger receptive fields. They are specifically tasked with handling complex targets that exhibit significant spectral variations or camouflage. A key component of our heavyweight experts is the integration of FFT-Attention, which allows for efficient global modeling.

Spectral-Domain Global Interaction. To prevent feature fragmentation caused by partition-based processing, we introduce a Spectral-Domain Interaction mechanism supplemented by a Shared Expert. As depicted in Figure 3, the system synergizes local spectral reasoning with global context. First, the selected complexity expert extracts features $\hat{x}_i^E$ using FFT-Attention. To balance efficiency and detail, this operation is performed within defined partitions of the input $x \in \mathbb{R}^{H \times W \times C}$. We project x into query Q and key K, transforming them into the frequency domain via 2D FFT ($\mathcal{F}$):

$$\hat{Q} = \mathcal{F}(W_q x), \quad \hat{K} = \mathcal{F}(W_k x). \tag{4}$$

The spectral interaction M_{spec} is computed via the Hadamard product ($\odot$) of the query and the key in the frequency domain. Unlike standard correlation, we perform element-wise multiplication without conjugation to model a convolutional interaction:

$$M_{spec} = \hat{Q} \odot \hat{K}. \tag{5}$$

The expert output $\hat{x}_i^E$ is reconstructed via the Inverse FFT ($\mathcal{F}^{-1}$) and modulated by the value V in the spatial domain:

$$\hat{x}_i^E = \mathcal{F}^{-1}(M_{spec}) \odot V + x. \tag{6}$$

Simultaneously, to compensate for the lack of global cross-window communication, we employ a Shared Expert S to capture holistic features. We adopt a Transposed Self-Attention (T-SA) module [El-Nouby *et al.*, 2021], which operates across the channel dimension:

$$S(x) = \text{T-SA}(x). \tag{7}$$

Finally, the output of the specific complexity expert is modulated by the holistic features from the shared expert. The final refined expert feature y_i^E is obtained as follows:

$$y_i^E = W_{out}\left(\hat{x}_i^E \odot \sigma(W_{mod} S(x))\right), \tag{8}$$

where σ denotes the SiLU activation, and W_{out}, W_{mod} are linear projections. This interaction ensures that even when a lightweight expert is selected, the output features retain necessary global spectral context.

3.3 Frequency Spectral Prompt Router

Effectively assigning inputs to the correct expert is critical. Relying solely on spatial semantics often fails to distinguish camouflaged targets. To address this, we propose the Frequency Spectral Prompt Router (FSP-Router). Instead of passive feature selection, the FSP-Router actively constructs spectral prompts from high-frequency cues to guide the routing decision toward discriminative regions.

Frequency-Aware Spectral Prompting. Spectral signatures of targets often exhibit distinct high-frequency variations (e.g., edges and textures) that differ from the background. We employ a fixed High-Pass Filter (HPF) in the spatial domain to extract these intrinsic cues. The input features x are processed by the HPF to obtain a high-frequency map x_{high}. This map is then projected to generate the spectral prompt E_{freq}. Unlike a simple feature channel, this prompt acts as an explicit indicator, highlighting regions with significant spectral-textural complexity to "prompt" the router's attention.

Prompt-Guided Routing. The routing function $g(x)$ assigns the input x to experts by aggregating spatial semantic features with the generated spectral prompt. The prompt functions as a learnable bias that modulates the routing logits. We adopt a Top-k selection mechanism:

$$g(x) = \text{topk}(\text{Softmax}(W_s x + \underbrace{W_f E_{freq}}_{\text{Spectral Prompt}} + \epsilon), k), \tag{9}$$

where W_s projects the spatial context, and W_f projects the spectral prompt. The prompt term shifts the decision boundary, encouraging the selection of higher-capacity experts for regions with complex high-frequency details. $\epsilon \sim \mathcal{N}(0, 1/n^2)$ is Gaussian noise.

Complexity Bias. To achieve an asymmetric load balance where simpler regions are routed to small-window experts, we incorporate a complexity bias in the auxiliary loss. We define the importance of an expert, $\text{Imp}_i(x)$, weighted by its parameter count p_i:

$$\text{Imp}_i(x) = \left(\sum_{x \in \mathbf{x}} \text{Softmax}(\cdot)_i\right) \cdot (p_i / p_{max}). \tag{10}$$

Experts with larger capacity (larger windows/depth) are logically biased to handle fewer, harder samples.

MoE Loss. The MoE loss is defined using the Coefficient of Variation (CV)[Riquelme *et al.*, 2021] of the importance (Imp) and load ($Load$) metrics:

$$L_{moe}(x) = \frac{1}{2} CV(\text{Imp}(x))^2 + \frac{1}{2} CV(\text{Load}(x))^2. \tag{11}$$

This ensures the router respects capacity constraints while avoiding collapse.

3.4 Loss Function

Our method adopts the OSTrack [Ye *et al.*, 2022] center point prediction method. Following OSTrack [Ye *et al.*, 2022], we use a weighted focal loss [Law and Deng, 2018]for classification and a combination of l1 loss and generalized IoU [Rezatofighi *et al.*, 2019] loss for regression. For task-recognition predictions, we use the cross-entropy loss as described earlier. The overall loss function is summarized as:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{G}}\mathcal{L}_{\text{IoU}} + \lambda_{L1}\mathcal{L}_{L1} + \mathcal{L}_{\text{moe}} \tag{12}$$

where L_{cls} denotes the weighted focal loss used for classification, L_{IoU} represents the generalized IoU loss, L_{L1} is the l1 regression loss, L_{moe} is defined in 3.3, and $\lambda_G = 2$ and $\lambda_{L1} = 5$ are the regularization parameters. For inference, we adopt a conventional template update strategy[Yan *et al.*, 2021], employing two templates: one static and one updated dynamically during tracking. The template update mechanism is governed by a straightforward approach, utilizing a fixed interval and a confidence threshold to determine when updates occur.

4 Experiments

4.1 Implementation Details

The SpecTrack models are implemented using Python 3.13 using PyTorch 2.9. Training is conducted on 8 NVIDIA RTX 3090 GPUs while inference speed is evaluated on a single NVIDIA RTX 3090 GPU.

Model. We develop five variants of SpecTrack models to strike a trade-off between speed and accuracy, each utilizing different transformer encoders and input resolutions, as detailed in Tab. 1.We adopt HiViT-L [Zhang *et al.*, 2023] as the transformer encoder for SpecTrack-L384 and L224, HiViTB for SpecTrack-B384 and B224, and HiViT-T for SpecTrack-T224. The transformer encoders are initialized with FastiTPN [Tian *et al.*, 2024] pre-trained parameters, which are further adapted to MSI-based vision tasks through a concise and effective interpolation-based reconfiguration strategy.The complete details and in-depth analysis of this strategy are provided in the appendix. In addition, we present the model parameters, FLOPs, and inference speed in Tab.1.

Table 1: Details of SpecTrack model variants.

Model	Transformer Encoder	Search Resolution	Template Resolution	Params (M)	FLOPs (G)	Speed (fps)
SpecTrack-L384	HiViT-L	384×384	192×192	434	289	6
SpecTrack-L224	HiViT-L	224×224	112×112	434	109	19
SpecTrack-B384	HiViT-B	384×384	192×192	117	81	17
SpecTrack-B224	HiViT-B	224×224	112×112	117	31	43
SpecTrack-T224	HiViT-T	224×224	112×112	30	9	80

Training. The training set comprises three standard benchmarks for multispectral object tracking, including MUST [Qin *et al.*, 2025],MSITrack [Feng *et al.*, 2025], HOT2022 [Xiong *et al.*, 2020b]and GOT10K [Huang *et al.*, 2021].The template and search images are generated by expanding the target bounding boxes by factors of 2 and 4, respectively. We adopt the AdamW optimizer with an initial learning rate of 10^{-4}. We train for 50 epochs with a batch size of 32 and reduce the learning rate by a factor of 10 after 30 epochs.We set $N = 4$ experts and Top-$k = 2$.

Inference. The online template is updated every 25 frames with a confidence threshold of 0.7. A Hanning window penalty is applied to incorporate positional priors, following standard practice [Ye *et al.*, 2022].

4.2 Evaluation Datasets and Metrics

We evaluate on four benchmarks. **MUST** [Qin *et al.*, 2025] is a large-scale UAV dataset with 250 sequences (43k frames) and 8 bands covering visible to near-infrared. **MSITrack** [Feng *et al.*, 2025] contains 300 videos (129k frames) with 8 bands (395–950 nm). **HOT2022** [Xiong *et al.*, 2020a] is a hyperspectral benchmark with 50 videos and 16 bands (470–620 nm). **GOT-10k** [Huang *et al.*, 2021] serves as a large-scale generic tracking benchmark.

We report standard metrics: Area Under the Curve (AUC), Success Rates at 0.5/0.75 ($SR_{0.5/0.75}$), Precision (P), and Normalized Precision (P_{norm}).

4.3 State-of-the-Art Comparisons

We evaluate our algorithm on three popular multispectral object tracking datasets and compare its performance with current state-of the-art methods.

Evaluation on MUST dataset. We evaluate SpecTrack on the large-scale MUST benchmark (250 sequences) in Tab. 2. Our variants consistently outperform existing methods. Notably, even SpecTrack-B224 surpasses previous state-of-the-art results. Specifically, SpecTrack-L384 outperforms the strongest prior UNTrack by 5.5% in AUC, 4.0% in SR0.5, and 14.8% in SR0.75, alongside gains of 3.4% and 4.7% in Pre and Pre$_N$. Additionally, as shown in Fig. 4, our method consistently excels across various challenging attributes compared to previous trackers. Fig. 5 visualizes that SpecTrack achieves tighter ground truth alignment than competitors in similar-color and cluttered-background cases.

Table 2: Comparison with state-of-the-art trackers on the MUST dataset. The top two results are highlighted in red and blue.

Method	AUC	SR$_{0.5}$	SR$_{0.75}$	Pre	Pre$_N$
OSTrack$_{256}$ [Ye *et al.*, 2022]	52.3	64.8	36.3	67.5	65.2
OSTrack$_{384}$ [Ye *et al.*, 2022]	55.1	69.6	44.1	73.3	68.8
ODTrack [Zheng *et al.*, 2024]	52.7	67.3	38.8	71.3	66.2
ZoomTrack [Kou *et al.*, 2023]	53.3	67.7	40.4	72.3	66.7
UNTrack [Qin *et al.*, 2025]	59.7	75.8	46.2	79.2	74.8
SpecTrack-T224	57.9	72.9	52.0	75.3	73.5
SpecTrack-B224	62.4	76.6	59.3	78.8	76.2
SpecTrack-B384	63.7	78.4	58.8	80.8	77.8
SpecTrack-L224	63.1	77.1	58.9	79.5	76.7
SpecTrack-L384	65.2	79.8	61.0	82.6	79.5

Evaluation on MSITrack dataset. We compare SpecTrack with state-of-the-art trackers on the MSITrack benchmark in

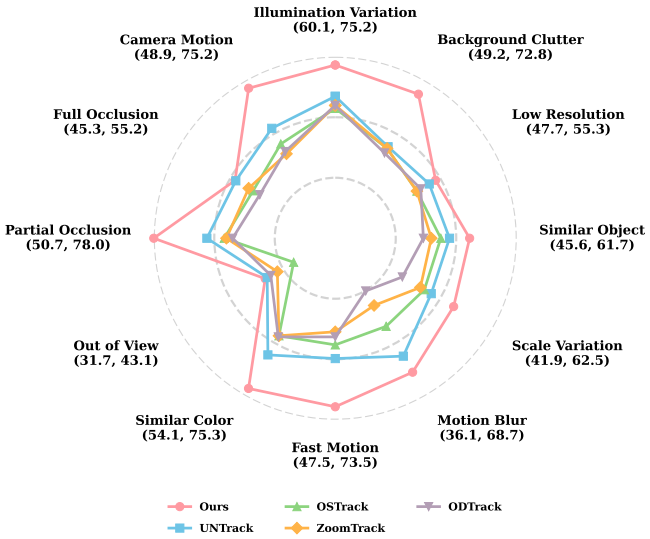

Figure 4: A comparison of SpecTrack with other state-of-the-art trackers in terms of AUC across each challenge attribute on MUST dataset, with the lowest and highest performance values marked.

Tab. 3. Our method consistently outperforms existing approaches. Notably, SpecTrack-L384 sets a new state-of-the-art, surpassing the strongest competitor GRM by 4.1% in AUC and 2.0% in Precision, with gains of 3.8% in $SR_{0.5}$ and 4.4% in Pre_N. Even the base variant, SpecTrack-B384, exceeds all prior methods in AUC and Pre_N, validating our framework's superiority in multispectral tracking.

Table 3: Comparison with state-of-the-art trackers on the MSITrack dataset. The top two results are highlighted in red and blue.

Method	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	Pre_N
$OSTrack_{256}$ [Ye *et al.*, 2022]	42.0	50.9	34.2	55.7	50.5
$OSTrack_{384}$ [Ye *et al.*, 2022]	43.1	52.5	36.1	56.2	52.4
LMTrack [Xu *et al.*, 2025]	43.2	52.4	39.7	55.2	52.5
ZoomTrack [Kou *et al.*, 2023]	44.3	54.7	35.1	59.0	54.4
NeighborTrack [Chen *et al.*, 2023b]	44.8	55.2	35.7	59.7	54.6
EVPTrack [Shi *et al.*, 2024]	46.2	55.8	43.9	58.3	56.0
AQATrack [Xie *et al.*, 2024]	47.1	56.7	44.7	58.5	56.9
UNTrack [Qin *et al.*, 2025]	47.3	58.4	40.5	61.8	57.8
GRM [Gao *et al.*, 2023]	47.8	58.1	40.8	63.0	57.8
SpecTrack-T224	43.3	52.7	40.7	55.1	52.2
SpecTrack-B224	48.4	58.6	43.3	63.2	58.2
SpecTrack-B384	49.8	61.4	45.8	63.8	60.8
SpecTrack-L224	50.7	61.1	46.4	63.5	60.7
SpecTrack-L384	51.9	62.2	48.9	65.0	62.2

Evaluation on HOT2022 dataset. We evaluate SpecTrack on the HOT2022 benchmark in Tab. 4. SpecTrack-L384 achieves state-of-the-art performance with 72.6% AUC and 95.0% Precision, outperforming previous best methods (HotMoE and UNTrack) by 2.0% in AUC. Regarding efficiency, SpecTrack-T224 runs at 80.0 FPS, while SpecTrack-B224 maintains competitive accuracy at 42.9 FPS. These results confirm that our approach effectively reduces computational redundancy while ensuring robust tracking.

Table 4: Comparison with state-of-the-art trackers on the HOT2022 dataset. The top two results are highlighted in red and blue.

Method	AUC	Pre	FPS
MHT [Xiong *et al.*, 2020a]	58.7	88.0	-
BAENet [Li *et al.*, 2020]	60.6	87.7	-
SSTNet [Li *et al.*, 2021]	62.1	90.1	-
SiamHYPER [Liu *et al.*, 2022]	67.8	94.5	27.7
SEENet [Li *et al.*, 2023]	65.3	93.0	13.1
HANet [Guo *et al.*, 2023]	68.8	94.8	21.2
HotMoE [Sun *et al.*, 2025]	70.4	93.5	43.7
UNTrack [Qin *et al.*, 2025]	70.4	93.7	37.0
SpecTrack-T224	63.8	91.2	80.0
SpecTrack-B224	70.2	93.4	42.9
SpecTrack-B384	71.2	95.0	17.2
SpecTrack-L224	72.4	95.1	19.0
SpecTrack-L384	72.6	95.0	6.0

Evaluation on GOT-10k dataset. To verify generalization, we evaluate SpecTrack on the RGB-based GOT-10k benchmark in Tab. 5. SpecTrack-L384 achieves top performance across all metrics, with 79.3% AO and 81.4% SR0.75, outperforming the previous best UNTrack by 2.0% in AO. Notably, even our base model SpecTrack-B224 (77.5% AO) surpasses UNTrack (77.3%). Furthermore, the significant 5.1% gain in SR0.75 over SPMTrack highlights our method's superior localization accuracy in general tracking scenarios.

Table 5: Comparison with state-of-the-art trackers on the GOT10K dataset. The top two results are highlighted in red and blue.

Method	AO	$SR_{0.5}$	$SR_{0.75}$
$OSTrack_{256}$ [Ye *et al.*, 2022]	71.0	80.4	68.2
$OSTrack_{384}$ [Ye *et al.*, 2022]	73.7	83.2	70.8
LoRAT [Lin *et al.*, 2024]	72.1	81.8	70.7
SeqTrack [Chen *et al.*, 2023a]	74.7	84.7	71.8
SPMTrack [Cai *et al.*, 2025]	76.5	85.9	76.3
UNTrack [Qin *et al.*, 2025]	77.3	88.4	74.7
SpecTrack-T224	72.9	82.2	71.0
SpecTrack-B224	77.5	87.1	78.2
SpecTrack-B384	78.9	87.8	79.6
SpecTrack-L224	78.5	87.5	79.4
SpecTrack-L384	79.3	88.7	81.4

4.4 Ablation Studies

In this section, we perform comprehensive ablation studies to validate the effectiveness of the proposed components in SpecTrack. All experiments are conducted using the SpecTrack-B224 model on the MUST dataset.

Table 6: Ablation study on the impact of SHMoE and FSP-Router components.

SHMoE	FSP-Router	AUC	Pre
		60.1	76.5
✓		61.3	77.4
✓	✓	**62.4**	**78.8**

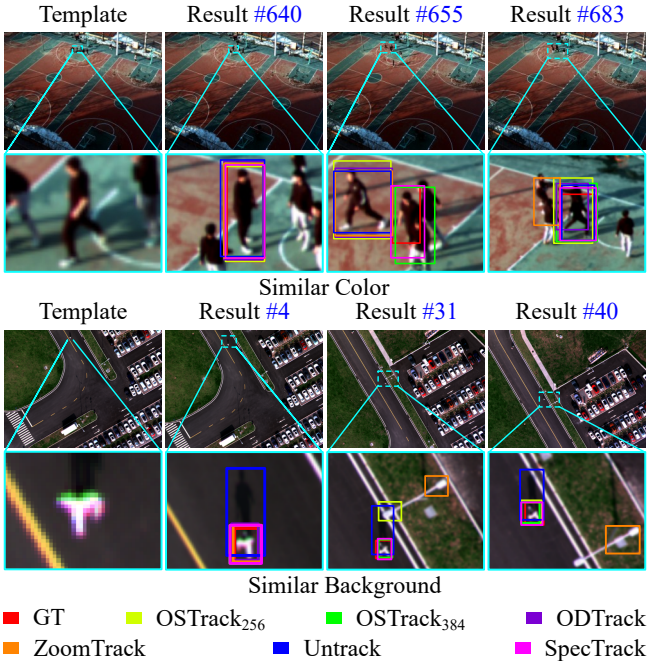

Figure 5: Visualization of our method with other multispectral trackers on four representative sequences that include multiple challenge attributes from MUST dataset.

Effectiveness of SHMoE Architecture. We first investigate the core contribution: the Spectral-Heterogeneous Mixture of Experts (SHMoE). As shown in Table 6, comparing the first two rows, introducing the SHMoE architecture alone yields a significant performance gain over the baseline. This confirms that the uniform processing paradigm is suboptimal for multispectral data. Qualitatively, as visualized in Figure 6, the baseline model tends to be distracted by background clutter, whereas the SHMoE-equipped model generates attention maps highly concentrated on the target, validating that heterogeneous experts effectively focus capacity on discriminative regions.

Impact of Spectral Heterogeneity. To verify the necessity of real multispectral information and the effectiveness of our adaptive design in exploiting spectral heterogeneity, we

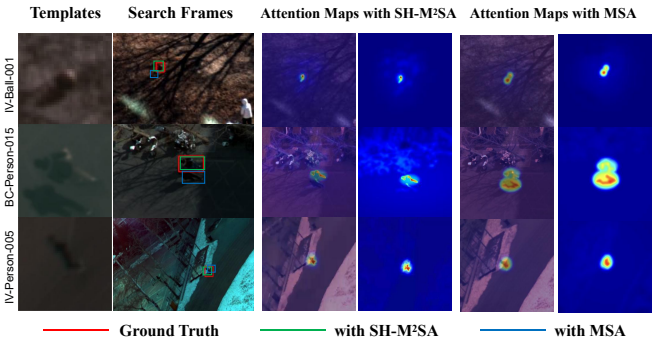

Figure 6: Visualization comparison of attention maps with MSA (Baseline) and with SH-M^{2}SA (Ours).

Table 7: **Ablation study on input modalities.** We compare the performance of the Baseline and our SpecTrack on RGB pseudo-color synthesized from MSI versus real MSI data.

Model	Input Data	AUC	Pre
Baseline	RGB	55.2	71.4
Baseline	MSI	60.1	76.5
SpecTrack	RGB	56.3	73.5
SpecTrack	MSI	**62.4**	**78.8**

Table 8: Ablation study on SHMoE hyperparameters: number of experts (N) and activated experts (k). The default setting is $N = 4, k = 2$.

N (Total)	k (Top-k)	AUC	Pre	FPS
3	2	61.7	77.6	**45.4**
4	2	62.4	**78.8**	42.9
5	2	**62.5**	**78.8**	39.7

conduct a comparative study between pseudo-color images (derived from 3-channel projections) and real Multispectral Imaging (MSI) data. The results are summarized in Table 7.

The results reveal two key insights. First, for both models, using real MSI data significantly outperforms using synthesized RGB, confirming that the rich spectral signatures in MSI provide critical discriminative cues lost in 3-channel projections. Second, SpecTrack achieves a much larger performance gain when switching from RGB to MSI compared to the Baseline. This demonstrates that our SHMoE architecture and FSP-Router effectively capture and leverage spectral heterogeneity, whereas the Baseline's uniform processing fails to fully exploit the complex spectral information.

Hyperparameters of SHMoE. We investigate the expert count (N) in Table 8. Results show that increasing N from 3 to 4 significantly enhances capacity, boosting AUC by 0.7%. However, further increasing N to 5 yields diminishing returns, resulting in only a marginal 0.1% gain in AUC. Meanwhile, the inference speed drops notably from 42.9 FPS to 39.7 FPS due to increased routing overhead. Consequently, we adopt $N = 4$ as the optimal configuration to strike the best balance between accuracy and efficiency.

5 Conclusion

In this work, we propose SpecTrack to address the inefficiency of uniform processing in multispectral tracking. By leveraging the Spectral-Heterogeneous Mixture of Experts (SHMoE) and Frequency Spectral Prompt Router, our framework dynamically aligns model capacity with scene complexity, achieving state-of-the-art performance on MUST and MSITrack, and verifies the scalability of the tracker on HOT2022 and GOT-10k.

Although effective in resolving spectral ambiguity, our current implementation shows suboptimal performance in full occlusion and out-of-view cases. Future work will be dedicated to exploring joint spatial-frequency-temporal modeling, aiming to leverage historical context for improved robustness in these challenging scenarios.

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